

Johnson C. Smith University

College Algebra | MTH 131(B) | 11-11:50AM SHB 203 | Fall 2013

"Every act of conscious learning requires the willingness to suffer an injury to one's self-esteem. That is why young children, before they are aware of their own self-importance, learn so easily; and why older persons, especially if vain or important, cannot learn at all."

—Thomas Szasz, author, professor of psychiatry (1920-)

Instructor Information

Name: Dr. James Ashe

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Office phone: (704) 378-1037

Office hours:

10:00 AM–11:00 AM MWF

12:00 PM–1:00 PM MW

12:00 PM–1:00 PM F (by appointment only)

1:00 PM–2:00 PM MW (by appointment only)

11:00 AM–12:00 PM TR (by appointment only)

Course Description

This is a 3 credit hour course designed to provide an investigative approach to college algebra using the computer and graphing calculator as tools. The topics covered include basic concepts of algebra, functions and graphs, polynomials and rational functions, exponential and logarithmic functions, systems of equations and solving systems of equations using matrices, sequences and series. At least one hour per week of computer assisted laboratory instruction is required. There are no prerequisites for this course.

Instructional methods will primarily consist of lectures and in-class discussions, however other methods such as collaborative projects may be adopted according to the particular needs of the class.

Course Objectives

Students should be able to demonstrate competency in the techniques and topics outlined in the section above. Additionally students should understand the relationship between an equation and its graph; be able to read, interpret, and solve application problems; and develop general problem solving skills.

Course Materials

Textbook/MyMathLab: A MyMathLab access codes is required. New textbooks and Ebooks usually come packaged with a MyMathLab code, *however you are only required to have a MyMathLab code.*

MyMathLab with the ebook *College Algebra*, 6th ed. Blitzer, Robert F. :

www.mypearsonstore.com/bookstore/mymathlab-with-pearson-etext-instant-access-for-college-0321782321

This is the text with MyMathLab code and ebook:

www.mypearsonstore.com/bookstore/college-algebra-9780321782281

Calculator: TI-83, TI-83+, and TI-84+ are supported by the department. Any other calculator model must be approved by the instructor.

Grades

Grades will be taken from homework, quizzes, and Exams as follows:

Evaluation: Attendance= 5%, Quizzes= 15%; Exams 1, 2, 3, 4= 10% each; Midterm and Final Exam = 20% each.

Grading scale: A $\geq$ 90 > B $\geq$ 80 > C $\geq$ 70 > D $\geq$ 60 > F

Attendance: 20% will be deducted from the attendance grade for every unexcused absence exceeding one.

Quizzes: Quizzes will consist of any graded assignment which is not an exam. This will include traditional quizzes, MyMathLab assignments, take home assignments, and in-class projects. The lowest quiz grade will be dropped.

Final Exam and Midterm: Both the final and midterm exam will be common comprehensive exams given in-class; see the exam schedule jcsu.edu/happenings/exam-schedule. If beneficial, the final exam score will replace the two lowest exam scores or the midterm.

Class Policies

JCSU policies regarding attendance, academic integrity, grades, and change of enrollment will be enforced. It is the student's responsibility to be aware of these policies and of related deadlines as well as any announcements or syllabus adjustments (written or oral) made in class. Please see the student handbook for details, jcsu.edu/docs/handbook.pdf.

Attendance: Class attendance is required for all JCSU students. Each student is allowed as many hours of absence per term as credit hours(s) received (not to exceed 4) for the class. Students who exceed the maximum number of absences may receive a failing grade for the course.

Students who may miss classes while representing the University in an official capacity are exempt from regulations governing absences. However, absence from class for official University business does not relieve the student from responsibility for any class assignments that may be missed during the period of absence.

Missed quizzes: There will be no make-ups for in-class quizzes unless the class was missed for a University sponsored event, and advance notice of the absence was given by the student. In this case the assignment will be excused or a make-up will be given depending on circumstances. Late take-home quizzes will have 25% deducted for each day it is late.

Missed exams: If a student misses an exam for a verifiable excuse, they should send me an e-mail *as soon as possible* so that a make-up can be arranged. Make-up exams may be significantly more difficult. If you will be gone for any reason, it will be *your* responsibility to inform me at least a week in advance to schedule to take the exam early.

Student support services: If you need course adaptations or accommodations for a documented disability please contact the Office of Disability Services.

jcsu.edu/directory/federal-requirements/general-institutional-information/jcsu-disability-services

phone: (704) 398-1116.

The Office of Counseling Services provides confidential intakes and assessments, personal counseling and referrals to appropriate resources to all enrolled students on a walk-in or appointment basis.

www.jcsu.edu/student/office-of-counseling-services, phone (704) 378-1044.

Course Content

Chapter 1: Equations and Inequalities

- 1.1 Graphs and Graphing Utilities
- 1.2 Linear Equations and Rational Equations
- 1.3 Models & Applications
- 1.4 Complex Numbers
- 1.5 Quadratic Functions
- 1.6 Other Types of Equations

Chapter 2: Functions and Graphs

- 2.1 Basics of Functions and Their Graphs
- 2.3 Linear Functions and Slope
- 2.4 More on Slope

Chapter 3: Polynomial and Rational Functions

- 3.1 Quadratic Functions
- 3.2 Polynomial Functions and Their Graphs

Chapter 4: Exponential and Logarithmic Functions

- 4.1 Exponential Functions
- 4.2 Logarithmic Functions
- 4.3 Properties of Logarithms

Important Dates

Friday 23 rd August	Last day to Add/Drop
Friday 6 th September	Exam 1
Wednesday 2 nd October	Exam 2
Wednesday 9 th October	Midterm Exam
Wednesday 30 th October	Exam 3
Friday 22 nd November	Exam 4
TBA	Common Final Exam

The above exam dates are subject to change. JCSU's Fall 2013 academic calendar can be found here: jcsu.edu/happenings/academic-calendar, and the exam schedule can be found here: jcsu.edu/happenings/exam-schedule.

While I do not foresee any significant changes to the items in this syllabus during the semester, I reserve the right to make appropriate changes. Students will be notified of any such changes in writing.